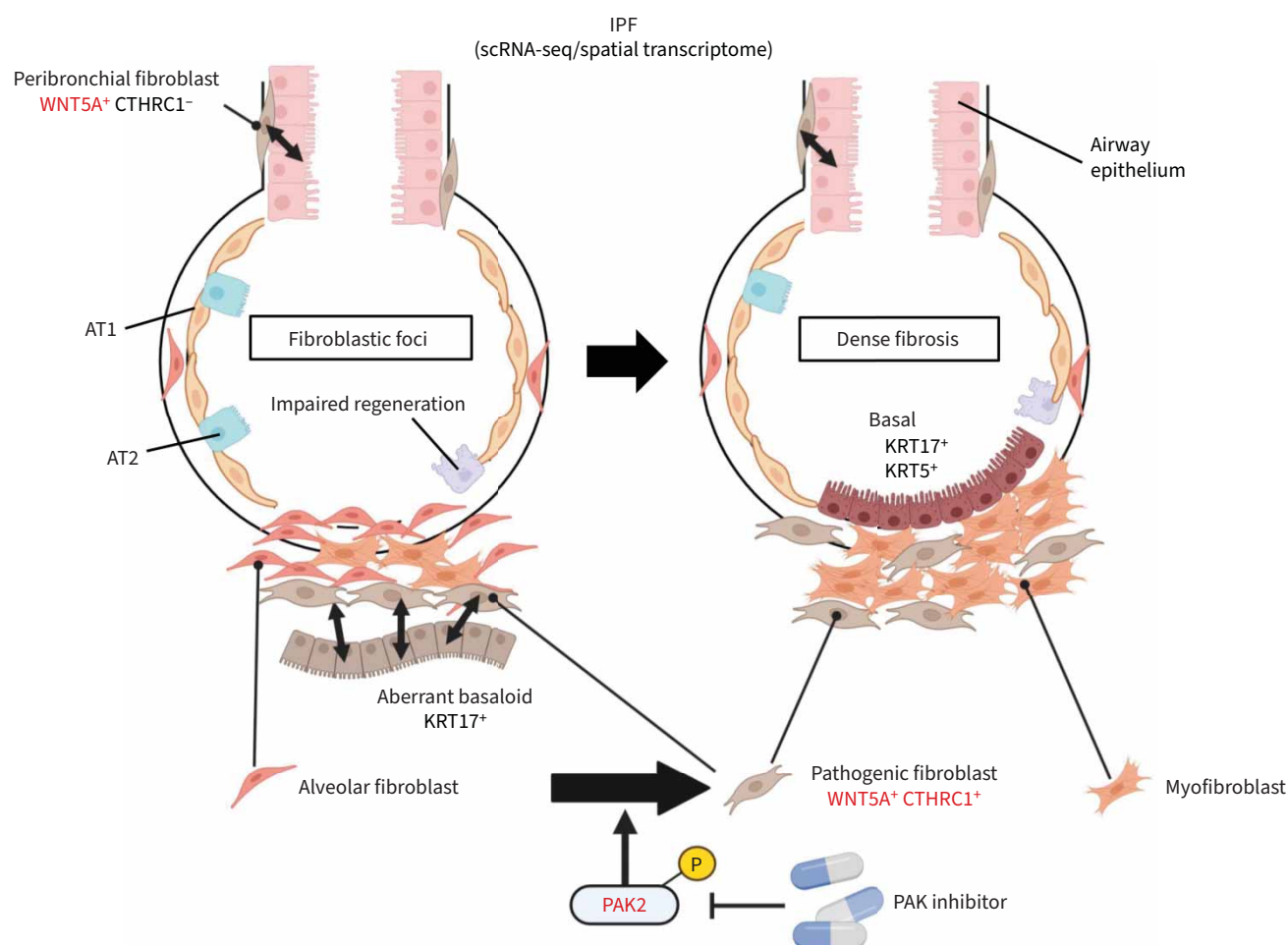


Integrated spatial and single-cell transcriptomics reveal PAK kinase as a therapeutic target in fibroblastic foci and dense fibrosis of idiopathic pulmonary fibrosis

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GRAPHICAL ABSTRACT Summary of the main study findings. Using integrated single-cell RNA sequencing (scRNA-seq) and spatial transcriptomics, we identified WNT5A⁺CTHRC1⁺ myofibroblasts enriched in both fibroblastic foci and dense fibrosis regions. Transcriptionally related to peribronchial fibroblasts, these cells aberrantly localise to the distal parenchyma in idiopathic pulmonary fibrosis (IPF) and closely interact with aberrant basaloid cells, forming a pro-fibrotic niche that promotes fibrosis progression. p21-activated kinase (PAK)2 was identified as an upstream regulator of these pathogenic fibroblasts, and its pharmacological inhibition attenuated their differentiation and fibrosis. AT1/2: alveolar type 1/2 cells. Created with BioRender.



Integrated spatial and single-cell transcriptomics reveal PAK kinase as a therapeutic target in fibroblastic foci and dense fibrosis of idiopathic pulmonary fibrosis

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Employing spatial transcriptomic techniques, this research has unveiled the importance of PAK kinase in the immature fibroblast niche during fibrogenesis, implying its potential as a therapeutic target for IPF <https://bit.ly/4kwlr73>

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Abstract

Idiopathic pulmonary fibrosis (IPF) is a fatal interstitial lung disease characterised by progressive fibrosis of lung parenchyma. The histopathology of IPF exhibits temporal and spatial heterogeneity, including immature fibroblastic foci (FF) and densely collagenised fibrosis. FF serve as dynamic niches of pro-fibrotic fibroblasts and play a pivotal role in fibrosis progression and transition into dense fibrosis (DF). Here, we integrated single-cell RNA sequencing (scRNA-seq) with spatial transcriptomics to elucidate cellular heterogeneity and the novel cell type involved not only in FF formation, but also in DF development. We identified a novel myofibroblast population, *WNT5A*⁺*CTHRC1*⁺ myofibroblasts, enriched in both FF and DF regions, underscoring their pivotal role in fibrosis progression. Differential gene expression analysis revealed the activation of p21-activated kinase (PAK)2 in these fibrotic areas, including *WNT5A*⁺*CTHRC1*⁺ myofibroblasts. PAK inhibition significantly suppressed transforming growth factor- β -induced myofibroblast differentiation and collagen production in IPF-derived fibroblasts. Furthermore, in a bleomycin-induced lung fibrosis mouse model, intraperitoneal administration of the PAK inhibitor significantly attenuated fibrotic progression. This study highlights the therapeutic potential of PAK inhibition for IPF, particularly targeting pathogenic fibroblasts within both FF and DF regions. By leveraging spatial transcriptomics and scRNA-seq, we provide a comprehensive molecular and cellular atlas of FF and DF in IPF lung tissue, offering new insights into fibrosis progression and therapeutic intervention.

Introduction

Idiopathic pulmonary fibrosis (IPF) is a chronic progressive interstitial lung disease of unknown aetiology. IPF shows a distinct histopathological pattern of usual interstitial pneumonia, with failure of alveolar re-epithelialisation, the persistence of fibroblasts/myofibroblasts, extracellular matrix (ECM) deposition, and architectural distortion, particularly in areas of dense fibrosis (DF) [1]. The areas between the fibrotic tissue and the morphologically normal lungs typically contain fibroblasts of variable sizes, termed



fibroblastic foci (FF) [2]. FF are the cardinal pathogenic lesions in IPF and consist of activated fibroblasts and myofibroblasts [3], the key effector cells responsible for a dysregulated ECM deposition at various fibrotic states. These structures represent the sites of myofibroblast accumulation during a pathological ECM deposition [4], consistent with the idea that fibrosis spreads from the FF into uninvolved alveoli. Indeed, a higher number of FF has been associated with worse outcomes in patients with IPF [5–8]. In contrast, DF represents areas of fibrotic remodelling, where mature scar tissue replaces normal lung architecture and severely impairs gas exchange [9–12]. Both FF and DF are critical components of IPF pathogenesis and potential therapeutic targets. However, the differences between FF and DF, as well as the fibroblast populations driving fibrosis in these regions, remain poorly understood.

Although bulk RNA sequencing has uncovered aspects of the IPF transcriptome, the mixture of FF, DF and normal tissue has limited the resolution of cell-type-specific insights. Single-cell RNA sequencing (scRNA-seq) has enabled the dissection of cellular heterogeneity in IPF [13–19], but lacks spatial context, which is essential given the regional nature of fibrogenesis. Spatial transcriptomics provides a means to localise gene expression, preserving architectural context [20]. Recent studies have utilised spatial transcriptomics to characterise FF at both cellular and molecular levels [21–23]. Despite these advances, the fibroblast populations associated with FF and DF and their respective roles in IPF progression remain incompletely characterised.

In this study, we integrated spatial transcriptomics and scRNA-seq in human IPF lungs to delineate the spatial distribution and transcriptional features of FF and DF. We identified a novel fibroblast subpopulation marked by *WNT5A* and *CTHRC1* enriched in both regions. Notably, we identified p21-activated kinase (PAK)2 as a critical fibrogenic kinase activated in FF and surrounding areas. Inhibition of PAK attenuated fibrosis in a murine model. These findings highlight the importance of PAK in the immature fibroblast niche during fibrogenesis, suggesting its potential as a therapeutic target for IPF.

Methods

Patient recruitment

All human lung tissue samples were obtained from patients who underwent lobectomy for local lung cancer at the Saitama Prefectural Cardiovascular and Respiratory Center (Japan). The IPF diagnosis for the patients included in this study was confirmed through a multidisciplinary discussion involving two board-certified pulmonary pathologists according to American Thoracic Society/European Respiratory Society guidelines [24]. No patients had received antifibrotic therapy. Clinicopathological details are provided in supplementary table S1.

Spatial transcriptomics, scRNA-seq and integrative analysis

Spatial transcriptomic analysis was performed using the Visium platform (10X Genomics) and sequenced on the DNBSEQ-G400 system (MGI). Data were processed with Space Ranger and analysed in Seurat with SCTransform normalisation. Cell-type deconvolution was performed *via* anchor-based integration with matched scRNA-seq data. Public datasets (GSE136831, GSE135893, GSE129605) and a published IPF cell atlas were integrated for cross-validation.

Fibroblastic foci index

The FF index was defined by the average expression of 20 fibrosis-related genes selectively upregulated in FF *versus* DF, identified using Seurat. Genes were ranked by expression specificity, excluding epithelial or immune-related markers.

Cell culture

Lung fibroblasts were isolated from histologically verified noncancerous tissue and cultured in DMEM+10% fetal bovine serum. Cells were transfected with 2 nmol·L⁻¹ short interfering (si)RNAs targeting PAK1/2 (Dharmacon) using Lipofectamine RNAi Max. RNA extraction, cDNA synthesis, quantitative (q)PCR and immunoblotting were performed using standard protocols. Primer and antibody details are provided in supplementary tables S9 and S10.

Immunostaining

Human and mouse lung sections were paraffin-embedded, stained (haematoxylin and eosin (H&E), trichrome, immunohistochemistry), and imaged using BZ-X710 microscopes. Antigen retrieval and antibody incubation followed standard protocols (supplementary table S10).

Animal studies

C57BL/6J mice received intratracheal bleomycin ($2.5 \text{ U} \cdot \text{kg}^{-1}$). FRAX486 ($60 \text{ mg} \cdot \text{kg}^{-1}$) or saline was intratracheally administered on days 7–20. On day 21, lung tissues were collected for histology and hydroxyproline assay.

Statistics

Welch's t-test, and ANOVA followed by Tukey's or Dunnett's test were used (GraphPad Prism version 9). Statistical significance was defined as $p < 0.05$.

Detailed methods are included in the supplementary material.

Results

Spatial transcriptomic analysis of human IPF lung tissue

Spatial transcriptomic profiling was performed on nine tissue sections from three patients with IPF using the Visium Spatial Gene Expression platform with freshly frozen lung resections (supplementary figure S1). After processing and quality control, each tissue section yielded an average of 2391 spots, with a median of 1789 genes expressed per spot (Methods, supplementary figure S1a, supplementary table S2). In addition, scRNA-seq was conducted on lung tissues from the same IPF patients and three healthy controls (figure 1a), revealing a skewed cell type composition in IPF lungs, consistent with previous findings (figure 1b and c, supplementary figure S1b–d).

To characterise the cellular and molecular features of FF, we identified them histologically in H&E images (figure 1d), and designated overlapping Visium spots as FF spots. FF spot selection was further validated by the spatial distribution of known FF markers, including *COL1A1*, *POSTN* and *CTHRC1*.

Detailed annotation of scRNA-seq data and its integration with spatial Visium spots

To further investigate the cell type composition within fibrotic regions of IPF, we refined the stromal cell type annotation using key markers implicated in IPF pathogenesis. Uniform manifold approximation and projection (UMAP) clustering identified nine distinct clusters (figure 2a and b), including *CTHRC1*⁺ fibroblasts, which are known to play a crucial role in IPF pathogenesis [25]. Notably, we identified an intermediate cell state bridging the fibroblast and smooth muscle cell (SMC)/pericyte clusters on the UMAP, which we termed *WNT5A*⁺*CTHRC1*⁺ myofibroblasts. These cells exhibited fibroblast-like characteristics, expressing *PDGFRA*, *PDGFRB* and *RUNX1*, while lacking *DES* expression, with unique markers such as *WNT5A* and *WIF1* (figure 2b, supplementary figure S2a). Proportions of myofibroblasts, *CTHRC1*⁺ fibroblasts, and *WNT5A*⁺*CTHRC1*⁺ myofibroblasts were enriched in IPF versus controls (figure 2c).

The presence of *WNT5A*⁺*CTHRC1*⁺ myofibroblast was validated in publicly available single-cell datasets from IPF patients (GSE136831, GSE135893), where *WNT5A*⁺ *WIF1*⁺ cells were detected within the myofibroblast cluster, mapped between the fibroblast and SMC clusters on UMAP (supplementary figure S2b and c). Although these cells shared transcriptional features with peribronchial fibroblasts in healthy controls [26], they exhibited distinct pro-fibrotic gene expression, particularly high levels of *POSTN*, *CTHRC1* and *LOXL1* (supplementary figure S2d) [17, 27, 28].

Pseudotime trajectory analysis revealed distinct fibroblast differentiation pathways in IPF compared to controls. In controls, fibroblasts predominantly follow a linear trajectory, transitioning into myofibroblasts. In IPF, alveolar fibroblasts diverged into two branches: one to myofibroblasts, the other to pathogenic *CTHRC1*⁺ fibroblasts, acquiring smooth muscle-like features (*TAGLN* and *ACTA2*), eventually giving rise to *WNT5A*⁺*CTHRC1*⁺ myofibroblasts (figure 2d and e, supplementary figure S2e). This bifurcation illustrates fibroblast heterogeneity in IPF and their roles in fibrogenesis. Immunohistochemistry of the human IPF lung tissue confirmed the expression of *WNT5A* and *CTHRC1* in FF and surrounding areas (figure 2f). Immunofluorescence co-staining further validated their presence in both FF and DF regions (figure 2g).

Molecular and cellular differences between FF and DF in human IPF lungs

Comparing FF with DF is crucial for understanding the dynamic progression of fibrosis and identifying therapeutic targets for both active fibrogenesis and its transition into fibrotic remodelling. We identified DF regions surrounding FF morphologically, characterised by fibroblast enrichment and the absence of SMCs and pericytes. Visium spots corresponding to DF regions were manually curated using deconvolution data from scRNA-seq to exclude spots with substantial SMCs or pericytes content (figure 3a). As a result, we identified 97 FF spots and 412 DF spots across nine sections. Principal component analysis demonstrated distinct transcriptomic profiles between FF and DF (figure 3b). Differentially expressed genes (DEGs) between FF and DF (supplementary table S4) are shown in a heatmap (supplementary figure S3a and b).

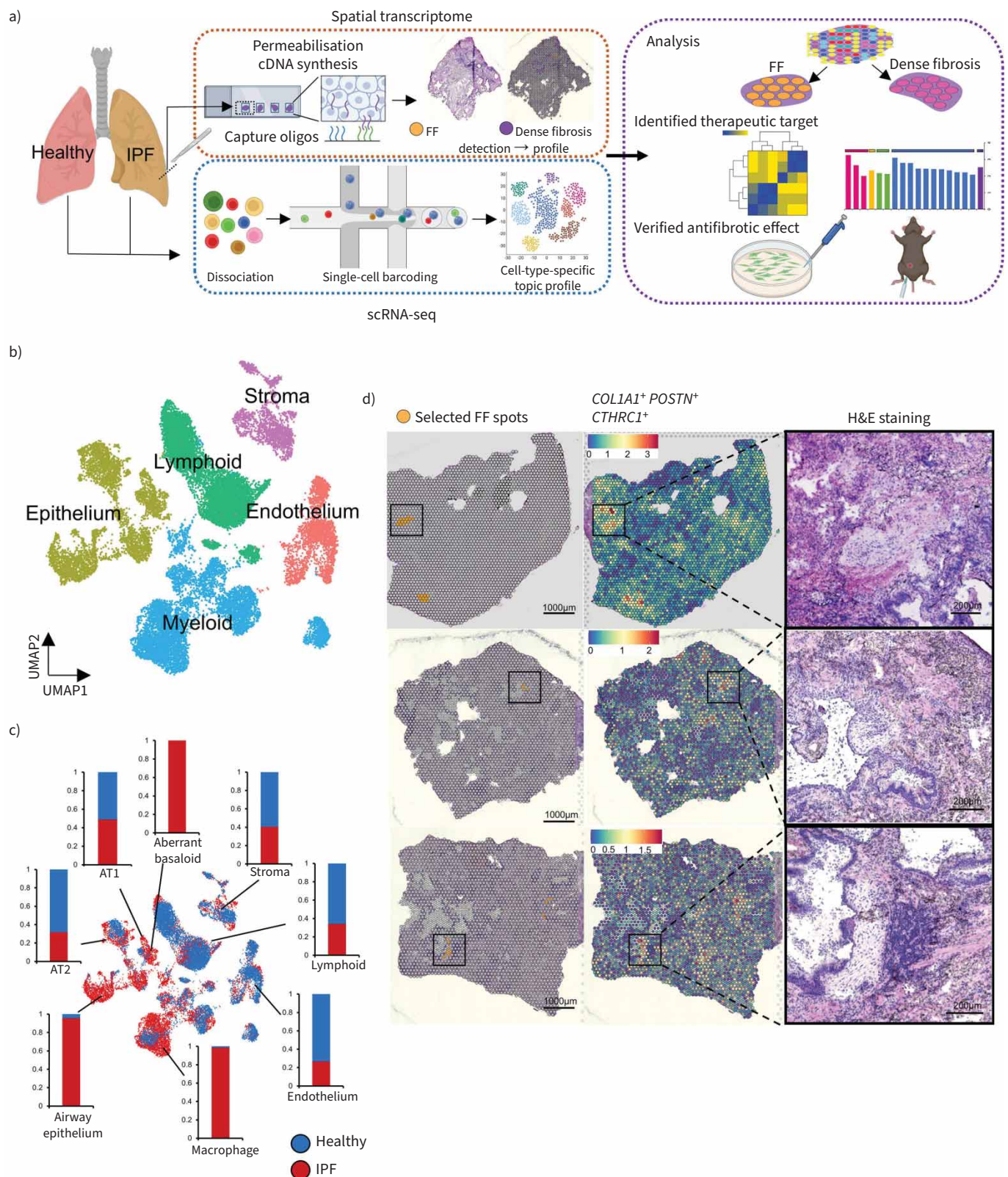


FIGURE 1 Study design and histological/molecular characterisation of idiopathic pulmonary fibrosis (IPF) lung tissue based on spatial transcriptome and single-cell RNA sequencing (scRNA-seq) analyses. **a)** Schematic workflow illustrating the experimental design for spatial transcriptome and scRNA-seq analysis of human IPF lungs. Nine lung sections from three IPF patients were used for spatial transcriptomic analysis, while three IPF and three control lung samples were used for scRNA-seq analysis. Detailed clinicopathological information is provided in supplementary table S1. **b)** Uniform manifold approximation and projection (UMAP) visualisation of the major cell types identified in the scRNA-seq

dataset, coloured by cell type. c) UMAP visualisation coloured by disease: bar plots comparing the relative proportions of key cell types in healthy control and IPF lung tissues. d) Spatial visualisation of fibroblastic foci (FF) annotations in representative sections, with expression levels of *COL1A1*, *POSTN* and *CTHRC1* shown. Representative zoomed-in images of the demarcated box area show haematoxylin and eosin (H&E) staining highlighting FF in IPF lung tissue. Scale bars=200 μ m. AT1/2: alveolar type 1/2 cells.

To elucidate the differences in cell type composition, deconvolution of FF and DF spots using stromal scRNA-seq data revealed distinct enrichment patterns: FF regions were predominantly populated by *CTHRC1*⁺ fibroblasts, proliferating fibroblasts and aberrant basaloid cells, while DF regions exhibited a higher prevalence of *WNT5A*⁺*CTHRC1*⁺ myofibroblasts, which were distinct from SMC/pericyte regions, further supporting the notion that transition to *WNT5A*⁺*CTHRC1*⁺ myofibroblasts can be partly responsible for the mechanisms for the DF development (figure 3c, supplementary figure S3c). Aberrant basaloid cells displayed high levels of cell–cell interactions with both *WNT5A*⁺*CTHRC1*⁺ myofibroblasts and *CTHRC1*⁺ fibroblasts, suggesting the presence of a pro-fibrotic amplification loop that drives fibrosis progression (supplementary figure S3d).

To molecularly detect fibroblast populations enriched in FF, we developed an FF index by identifying FF-specific transcriptional markers through differential gene expression analysis between FF and DF regions (Methods, supplementary figure S3b, supplementary table S5). Although fibroblasts with a high FF index were also detected in control lung samples, the amount of fibroblasts with a high FF index was higher in the IPF samples (supplementary figure S3e). Spatially, the FF index effectively visualised FF in the IPF lung sections (figure 3d). Fibroblasts with a high FF index were predominantly composed of pathogenic fibroblasts, including *CTHRC1*⁺ fibroblasts, proliferating fibroblasts and *WNT5A*⁺*CTHRC1*⁺ myofibroblasts, while alveolar fibroblasts were more prevalent in the fibroblasts with a low FF index (figure 3e). These results indicate that high-FF-index fibroblasts represent pathogenic populations and potential therapeutic targets.

Identification of PAK2 activation in FF and surrounding dense fibrosis area and its therapeutic potential in IPF

To identify novel therapeutic targets for IPF, we further analysed DEGs between FF and DF. Ingenuity Pathway Analysis (IPA) revealed activation of the pulmonary fibrosis idiopathic signalling pathway and upstream regulators including several kinases in FF (figure 3f, supplementary tables S6 and S7). Among these kinases, *PRKAA2*, a catalytic subunit of AMP-activated protein kinase (AMPK), emerged as a candidate due to its potential role in modulating transforming growth factor (TGF)- β -induced myofibroblast differentiation. We have previously reported the antifibrotic effects of metformin targeting AMPK signalling [29]. As part of our search for novel therapeutic targets, we focused on PAK2, given its high expression in pathogenic fibroblasts and availability of small-molecule inhibitors (figure 3g).

To assess PAK2 activity, we established a PAK target gene set (supplementary table S8) based on the previously reported RNA-seq data [30]. The average expression level of the PAK target gene signature in each single cell was calculated using the module score algorithm and visualised on the UMAP (figure 4a). The distribution of cells with high PAK target signature expression closely aligns with cells having a high FF index, particularly within the IPF fibroblast population (figure 4b and c). This suggests that PAK target genes are predominantly expressed in pro-fibrotic fibroblasts. Furthermore, PAK2 expression was significantly higher in fibroblasts with a high FF index compared to those with a low FF index, as shown in the violin plot (figure 4d). Spatial distribution analysis in the Visium dataset revealed that PAK target gene expression is enriched around FF and DF regions. This enrichment closely corresponds to the distribution of pathogenic fibroblast populations, including *CTHRC1*⁺ fibroblasts in FF and *WNT5A*⁺*CTHRC1*⁺ myofibroblasts in DF, as shown in spatial mapping (figure 4e). These findings support the association of PAK target gene activity with pro-fibrotic microenvironments in IPF lungs.

Among the PAK family, PAK2 showed the highest expression compared to PAK1 and PAK3, with significant enrichment in IPF lung tissue at both the transcriptome and protein levels (figure 4f and g). These findings underscore the potential of PAK2 as a therapeutic target in IPF by modulating pathogenic fibroblast activity.

The role of PAK2 in TGF- β induced pro-fibrotic differentiation in human lung fibroblast and its inhibition as a therapeutic target

Using freshly isolated primary human lung fibroblasts, we performed siRNA-mediated knockdown of PAK1 and PAK2 to elucidate their role in TGF- β induced pro-fibrotic properties. Both PAK1 and PAK2

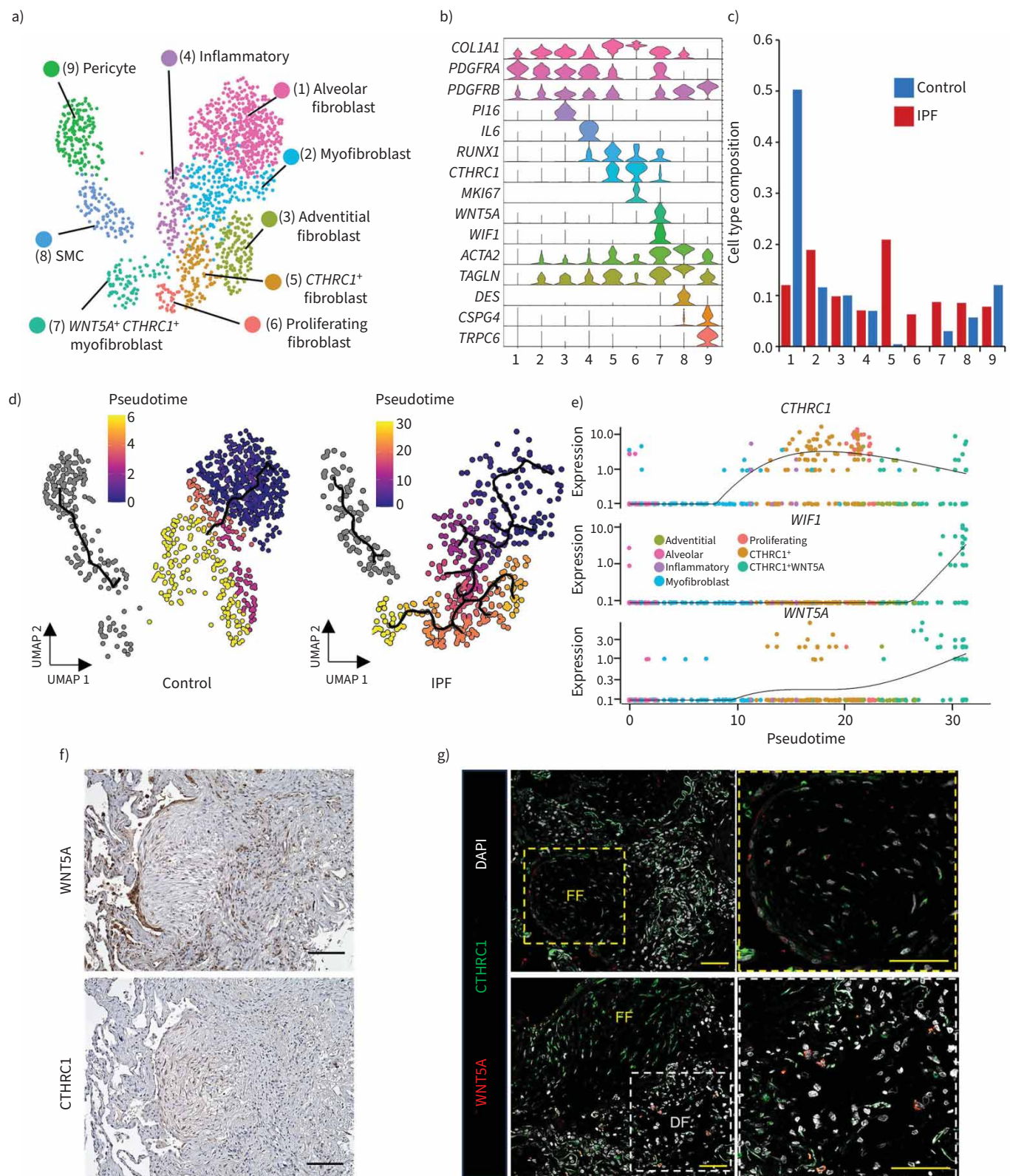


FIGURE 2 Single-cell RNA sequencing (scRNA-seq) analysis of stromal cell populations reveals distinct cell type composition and differentiation trajectory in human idiopathic pulmonary fibrosis (IPF) lung. **a)** Uniform manifold approximation and projection (UMAP) visualisation of the annotated stromal scRNA-seq dataset from three IPF and three control lung samples, labelled by detailed cell type. **b)** Violin plot illustrating the expression of selected marker genes in each cluster. **c)** Bar plot comparing the composition of lung stromal cell types between normal and IPF patients. **d)** Pseudotime developmental trajectory analysis generated using Monocle3, demonstrating distinct stromal cell differentiation pathways

in healthy control *versus* IPF lungs. **e)** Expression trends of *CTHRC1*, *WIF1* and *WNT5A* along the pseudotime trajectory in IPF lungs, showing their progressive upregulation during the transition to pro-fibrotic states. **f)** Immunohistochemical staining of *WNT5A* and *CTHRC1* in fibroblastic foci (FF) and surrounding fibrotic regions of IPF lung tissue. Scale bars=100 μ m. **g)** Immunofluorescence staining of *WNT5A* and *CTHRC1* in FF and dense fibrosis (DF) regions of IPF lung tissue, showing co-localisation in pathogenic fibroblasts. Nuclei are counterstained with 4',6-diamidino-2-phenylindole (DAPI). Scale bars=50 μ m. SMC: smooth muscle cell.

knockdown efficiently reduced TGF- β induced ECM production and *CTHRC1*, *WNT5A* expression (figure 5a and b).

To characterise the pharmacological antifibrotic properties targeting PAK2 in terms of IPF treatment, we tested two PAK inhibitors, FRAX486 and FRAX597 in TGF- β -induced myofibroblast differentiation (figure 5c). While FRAX486 and FRAX597 inhibit multiple PAK family members, including PAK1, PAK2 and PAK3 [31, 32], their primary effect is probably mediated through PAK2, given its expression pattern in IPF lung tissue (figure 4f). Both inhibitors suppressed TGF- β -induced α -smooth muscle actin (SMA) and type I collagen (figure 5c), in a dose-dependent manner (figure 5d), outperforming pirfenidone and nintedanib. Immunofluorescence confirmed the reduced protein expression of α -SMA and type I collagen (supplementary figure S4a). To examine downstream effects of PAK inhibition, we analysed protein expression using Western blotting (figure 5e, supplementary figure S4b). Both inhibitors suppressed the expression and phosphorylation of PAK2, c-Abl and LIMK1, but not SMAD2/3. In particular, FRAX486 strongly inhibited TGF- β -induced myofibroblast differentiation in lung fibroblasts. The transcriptional inhibition of downstream factors during TGF- β -induced myofibroblast differentiation was confirmed using qPCR (supplementary figure S4c).

In IPF-derived fibroblasts, FRAX486 and FRAX597 also reduced α -SMA and type I collagen expression (figure 5f and g). PAK2 downstream signalling were similarly suppressed (supplementary figure S4d–f). Taken together, these findings suggest that the antifibrotic effects of FRAX486 and FRAX597 are mediated through the suppression of pro-fibrotic differentiation in both TGF- β -induced models and IPF-derived lung fibroblasts.

Therapeutic potential of the PAK2 inhibitors in an *in vivo* fibrosis model

To investigate the physiological antifibrotic effects of FRAX486 and FRAX597 *in vivo*, we employed a bleomycin (BLM)-induced lung fibrosis mouse model (figure 6a). Analysis of published scRNA-seq datasets [7] showed elevated PAK2 expression, but not PAK1/PAK3, in fibroblasts from BLM-treated lungs (figure 6b). Additionally, we confirmed enhanced phosphorylation of PAK2 in lung tissues of the BLM-induced fibrosis model, indicating its activation during fibrotic progression (figure 6c).

Based on *in vitro* results showing greater efficacy of FRAX486 over FRAX597, we selected FRAX486 for *in vivo* analysis. FRAX486 was administered intraperitoneally on day 7 after BLM treatment, corresponding to the onset of the fibrotic phase, to evaluate its therapeutic potential (figure 6a). As shown in figure 6d, BLM-treated mice exhibited significant weight loss, which was restored by FRAX486 treatment to levels comparable to controls by day 21.

Histological analyses confirmed the antifibrotic effects of FRAX486. A hydroxyproline assay and Masson's trichrome staining demonstrated significant attenuation of collagen deposition and fibrosis (figures 6e and f). Ashcroft scoring also revealed a substantial reduction in fibrotic lesions (figure 6g). Immunohistochemistry showed that FRAX486 significantly inhibited phosphorylated (p-)PAK2 expression in the BLM-treated lungs (figure 6h), consistent with its proposed mechanism of action. Notably, p-PAK2 was more prominently induced by BLM treatment compared to p-PAK1, highlighting the central role of PAK2 activation in fibrosis (figure 6h). Immunofluorescence co-staining further demonstrated that FRAX486 markedly reduced *WNT5A*⁺*CTHRC1*⁺ fibroblasts in BLM lungs (supplementary figure 5a and b).

These findings collectively demonstrate that FRAX486 mitigates PAK2-mediated signalling and reduces fibrosis severity in a BLM-induced lung fibrosis model, supporting its potential as a therapeutic agent for IPF.

Discussion

IPF is characterised by spatially and temporally heterogeneous fibrosis involving distinct fibroblast populations in FF and DF. FF are dynamic niches of activated fibroblasts and myofibroblasts responsible for dysregulated ECM deposition. DF represents advanced fibrotic remodelling characterised by severe

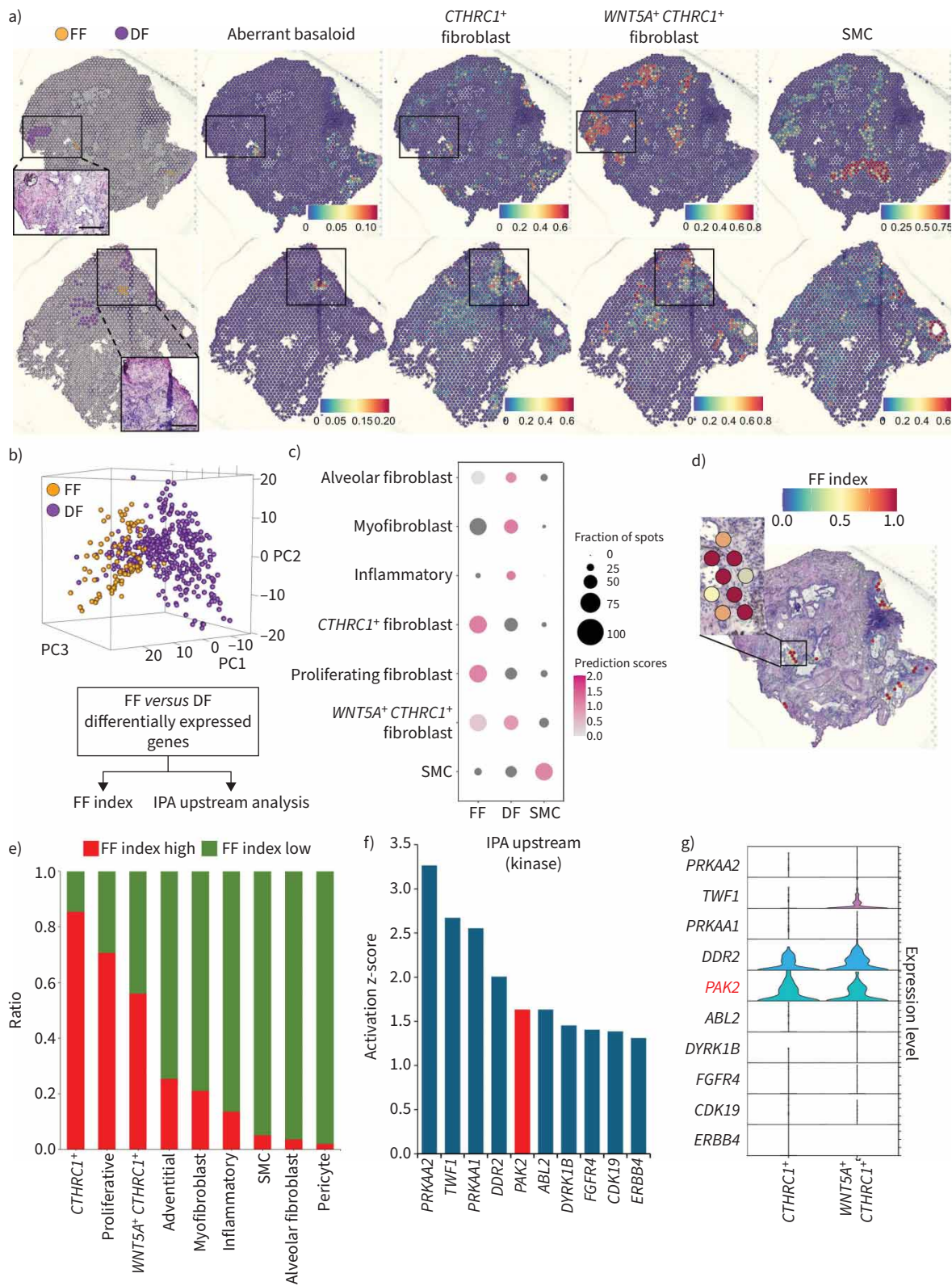


FIGURE 3 Comparative cellular and molecular analysis between the fibroblastic foci (FF) and dense fibrosis (DF) in human idiopathic pulmonary fibrosis (IPF) lungs. **a)** Spatial visualisation of DF annotations alongside inferred cell-type distributions on representative Visium sections. Specific cell types, including aberrant basaloid cells, *CTHRC1*⁺ fibroblasts, *WNT5A*⁺*CTHRC1*⁺ myofibroblasts and smooth muscle cells (SMCs), are displayed with heatmaps of their predicted abundance. Representative zoomed-in images of the demarcated box area showing haematoxylin and eosin (H&E) staining, highlighting FF and DF in IPF lung tissue. Scale bar=500 μm. **b)** Three-dimensional principal component (PC) analysis mapping with 97 FF

spots and 412 DF spots. **c**) Dot plot showing inferred cell-type densities for selected stromal cell types, comparing FF, DF and SMC regions. **d**) Spatial distribution of the FF index on a representative Visium section (patient JKPF2-1). The FF index was calculated using the average expression levels of 20 fibrosis-related genes, listed in supplementary table S5. **e**) Bar plot showing the proportion of stromal cell types with high and low FF index populations. **f**) Upstream regulator prediction of 746 differentially expressed genes between FF and DF regions, based on Ingenuity Pathway Analysis (IPA). The top 10 upstream regulators predicted as “activated” are shown. **g**) Violin plots displaying the average expression levels of the top 10 upstream regulators identified in **f**), in each cell type.

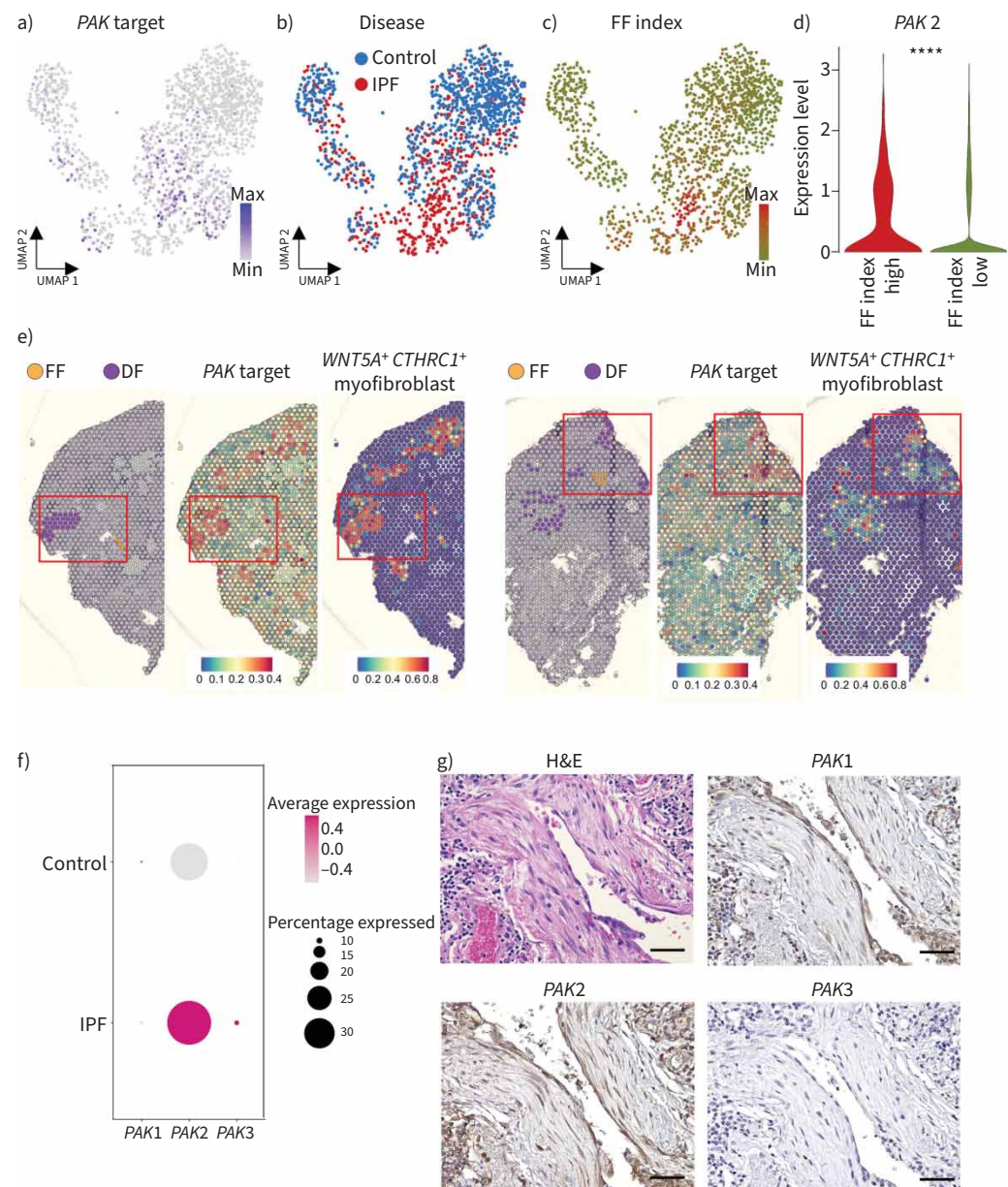


FIGURE 4 Molecular and spatial analysis of p21-activated kinase (PAK) target genes in human idiopathic pulmonary fibrosis (IPF) lung tissue. **a**) Visualisation of PAK target gene signature; **b**) disease status; and **c**) fibroblastic foci (FF) index displayed on the uniform manifold approximation and projection (UMAP). **d**) Violin plot showing PAK2 expression in stroma cells with a high FF index compared to those with a low FF index. **e**) Spatial visualisation of PAK target gene signature alongside FF/dense fibrosis (DF) annotations and inferred cell-type distributions on representative Visium sections. **f**) Dot plot showing the expression of PAK family genes (*PAK1*, *PAK2*, *PAK3*) within stromal cells in control and IPF lung samples. **g**) Representative haematoxylin and eosin (H&E) staining and immunohistochemical analysis of *PAK1*, *PAK2* and *PAK3* in fibroblastic foci of IPF lung tissue. Scale bars=50 µm. ****: p<0.0001.

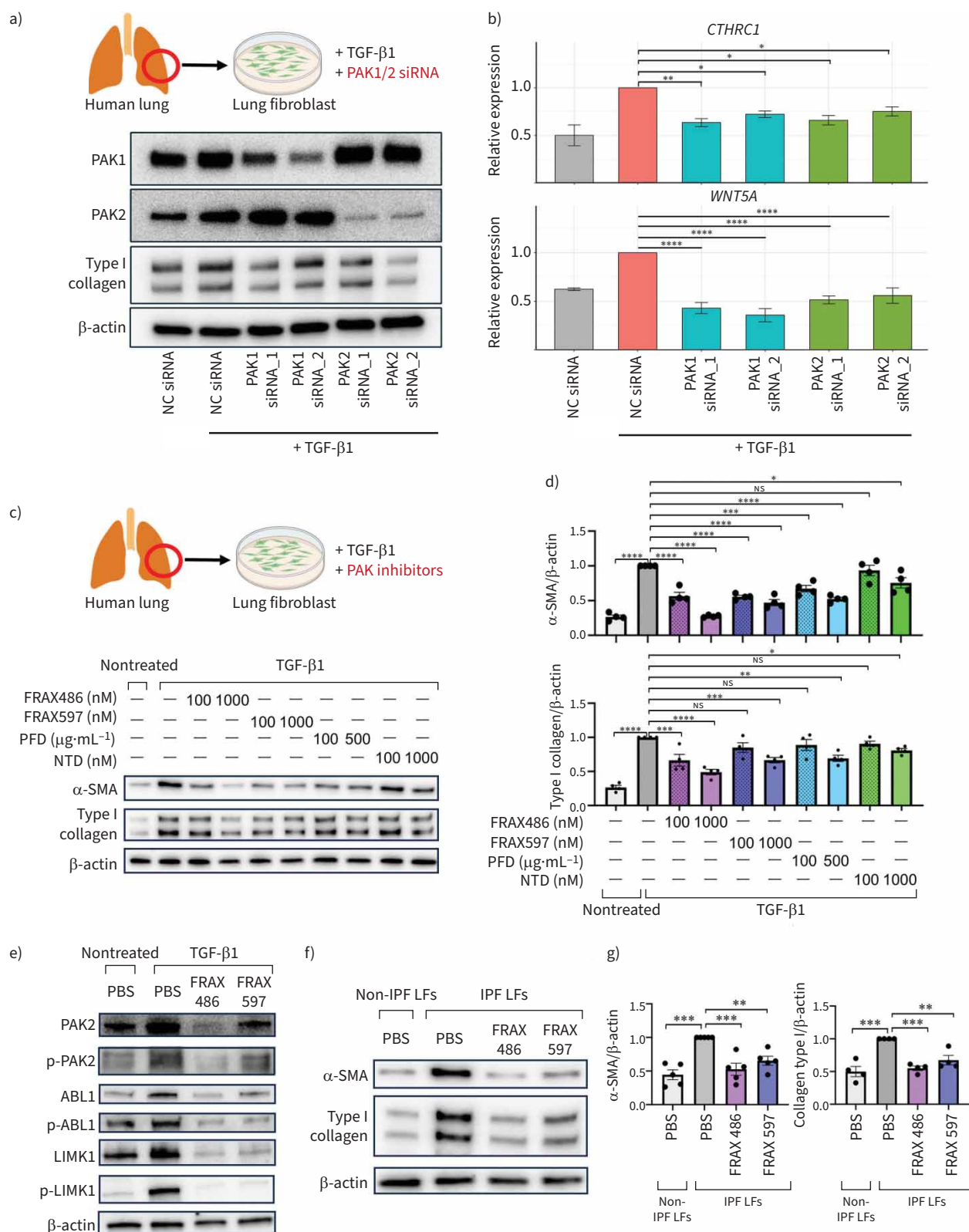


FIGURE 5 p21-activated kinase (PAK) inhibition alleviates transforming growth factor (TGF)- β -induced or idiopathic pulmonary fibrosis (IPF) lung myofibroblast differentiation. **a)** The effect of the short interfering (si)RNA-mediated PAK1/2 knockdown on the TGF- β -induced myofibroblast differentiation of lung fibroblasts. Schematic protocol illustrating the *in vitro* experiment. Representative immunoblot analysis showing the levels of PAK1, PAK2, type I collagen and β -actin in the PAK1 or PAK2 knockdown lung fibroblasts after 48 h TGF- β 1 (2 ng·mL⁻¹) treatment. **b)** Quantitative (q)PCR showing the expression levels of *CTHRC1* and *WNT5A* in the PAK1 or PAK2 knockdown human lung fibroblast after 48 h TGF- β 1 (2 ng·mL⁻¹)

treatment. **c)** The effect of the PAK2 inhibitors FRAX486 and FRAX597 on the TGF- β -induced myofibroblast differentiation of lung fibroblasts. Schematic protocol illustrating the *in vitro* experiment. Representative immunoblot analysis showing the amount of α -smooth muscle actin (SMA), type I collagen and β -actin in the lung fibroblasts treated for 48 h with FRAX486 (100 nM and 1000 nM) and FRAX597 (100 nM and 1000 nM) in the presence of TGF- β 1 (2 ng·mL⁻¹). Pirfenidone (PFD) and nintedanib (NTD) were used as positive controls to inhibit myofibroblast differentiation. **d)** Quantitative immunoblot analysis showing the amount of type I collagen and α -SMA. **e)** Immunoblot analysis showing the amount of PAK2 and its downstream signals in TGF- β -induced myofibroblasts treated with FRAX486 (1000 nM) and FRAX597 (1000 nM) in the presence of TGF- β 1 for 48 h. **f)** The effect of PAK2 inhibitors on IPF lung fibroblasts. Representative immunoblot analysis showing the amount of α -SMA, type I collagen and β -actin in IPF lung fibroblasts treated with FRAX486 (1000 nM) and FRAX597 (1000 nM) for 48 h. Protein samples were collected from treated cells 48 h after initiating the treatment. **g)** Quantitative immunoblot analysis showing the amount of type I collagen and α -SMA in IPF lung fibroblasts. NC: negative control; NS: nonsignificant. *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$.

architectural disruption. Understanding the cellular composition and molecular mechanisms underlying FF and DF is crucial for identifying novel therapeutic strategies.

The complementary strengths of scRNA-seq and spatial transcriptomics make their integration a powerful approach to simultaneously reveal the heterogeneity of fibrosis and the spatial distribution of diverse cellular microenvironments. By combining these two technologies, our study provides a detailed spatial and molecular atlas of the IPF lung, focusing on the fibroblast populations within FF and DF. This approach revealed critical pro-fibrotic pathways and highlighted the role of specific fibroblast subpopulations in fibrosis progression. Notably, FF is enriched with *CTHRC1*⁺ fibroblasts and proliferating fibroblasts, whereas DF shows an increased proportion of myofibroblasts, including *WNT5A*⁺*CTHRC1*⁺ myofibroblasts. Among these, the identification of *WNT5A*⁺*CTHRC1*⁺ myofibroblasts as a distinct pro-fibrotic subpopulation marks a novel contribution of this study. Interestingly, the transcriptional profiles of *WNT5A*⁺*CTHRC1*⁺ myofibroblasts show some resemblance to peribronchial fibroblasts in healthy controls, suggesting that parenchymal fibroblasts may adapt to the fibrotic microenvironment. Additionally, *WNT5A*⁺*CTHRC1*⁺ myofibroblasts interact dynamically with aberrant basaloid cells, potentially driving a pro-fibrotic amplification loop. This proximal-like adaptation in distal airspace could partially explain the remodelling processes in fibrotic lungs.

Recent studies have demonstrated that WNT5A, a key noncanonical WNT ligand, promotes TGF- β via JNK and ROCK signalling pathways, contributing to fibroblast-to-myofibroblast transition [33]. Our findings extend these insights by demonstrating that *WNT5A*⁺*CTHRC1*⁺ myofibroblasts, probably derived from *CTHRC1*⁺ fibroblasts in FF, are enriched in both FF and DF regions. Their presence across these regions underscores their potential role in bridging early fibrotic activity in FF with advanced remodelling processes in DF. However, the regulatory mechanisms linking these stages of fibrosis remain to be clarified.

Notably, this integrated spatial transcriptomic methodology enabled the identification of PAK2, as a key regulatory kinase highly expressed in pathogenic fibroblasts, including *CTHRC1*⁺ fibroblast and *WNT5A*⁺*CTHRC1*⁺ myofibroblast. The PAK family consists of serine/threonine kinases that regulate diverse cellular activities [34]. Among its six isoforms (PAK1–6), PAK2 is widely expressed and functions as a critical effector of the Rho family of small GTPases, linking it to cytoskeletal reorganisation [35]. PAK2 also plays roles in tumorigenesis, cellular senescence and ageing [36, 37]. In fibrosis, PAK2 functions downstream of TGF- β and AKT, and is essential for TGF- β -mediated pro-fibrotic signalling [38], serving as a noncanonical SMAD effector [39–41]. While PAK2 has been implicated in hepatic stellate cells [42] and peritoneal mesothelial cells [43] and renal fibrosis [41], its role in IPF remains poorly defined.

In contrast, PAK1 has been studied as a regulator of myofibroblast mechanotransduction in liver and lung fibrosis. Its silencing or inhibition suppresses pro-fibrotic phenotypes *in vitro* and *in vivo*, including in the bleomycin-induced lung fibrosis mouse model [44], consistent with our *in vitro* findings. However, our scRNA-seq and immunohistochemistry data indicate that PAK1 expression is relatively weak in mesenchymal cells of the IPF lung, compared to the predominant expression of PAK2 in fibroblastic foci and surrounding regions (figure 4f). Importantly, PAK2, but not PAK1, was identified in the IPA analysis as a key upstream regulator in fibroblastic regions, suggesting a differential contribution of PAK isoforms to IPF pathogenesis. Spatially, *CTHRC1*⁺ fibroblast and *WNT5A*⁺*CTHRC1*⁺ myofibroblast, which exclusively expresses PAK2 within stromal populations, closely aligns with FF and DF regions. While the nonspecific effects of FRAX486, particularly on PAK1, employed in our *in vivo* model present a limitation in interpreting our findings, our integrated spatial and single-cell transcriptomics analysis support the

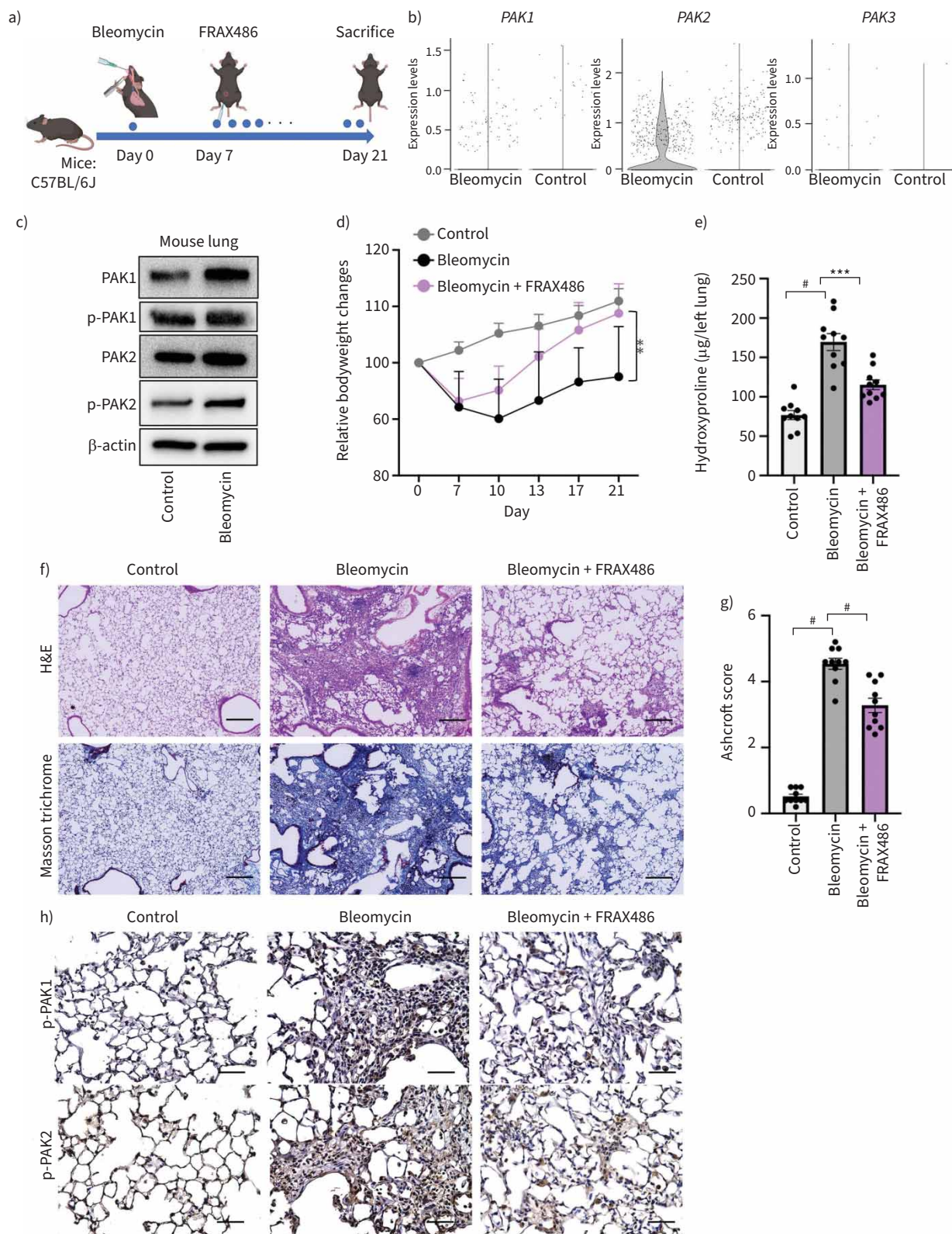


FIGURE 6 The administration of the p21-activated kinase (PAK)2 inhibitor FRAX486 attenuates bleomycin-induced lung fibrosis in mice. **a)** Schematic protocol used for the FRAX486 treatment of the mice with bleomycin-induced pulmonary fibrosis. **b)** Violin plots showing the expression levels of *Pak1*, *Pak2* and *Pak3* in fibroblasts isolated from bleomycin-treated and control mouse lungs (GSE129605). **c)** Representative immunoblot showing PAK1, p-PAK1, PAK2 and p-PAK2 levels in lung tissue lysates from bleomycin-treated and control mouse lungs. **d)** Bodyweight changes observed after the bleomycin treatment. The bodyweight at day 0 before initiating the treatment was attributed a relative value of 100.

Data are presented as mean $\pm$ SEM. Control n=10; bleomycin control n=10; bleomycin+FRAX486 n=10. **e)** Quantification of hydroxyproline in each left lung of the mice. Each dot represents the data from one animal. **f)** Haematoxylin and eosin (H&E) and Masson's trichrome staining of representative lung sections from each indicated group of treated mice. Scale bars=100 μ m. **g)** Quantification of fibrosis using the Ashcroft score. Each dot represents the data from one animal. **h)** Immunohistochemical staining of phosphorylated (p-)PAK1 and phosphorylated p-PAK2 in lung tissue sections from each treatment group. Scale bars=50 μ m. **: p<0.01, ***: p<0.001, #: p<0.0005.

specific role of PAK2 in the pro-fibrotic fibroblast populations. Further *in vivo* studies comparing PAK1 and PAK2 knockdown, and use of PAK2-specific inhibitors, are warranted to validate this specificity.

We acknowledge the potential limitations in defining DF due to the fundamental resolution constraints of the Visium platform. To mitigate this, we combined histological examination with scRNA-seq-based deconvolution to exclude smooth-muscle-rich regions and focus on myofibroblast-enriched areas. Nevertheless, spot-level data may overlook cellular and molecular heterogeneity within these regions.

In conclusion, our integrative approach illuminates spatially resolved fibroblast subtypes underlying IPF progression. By targeting both FF and DF-associated fibroblasts, PAK2 inhibition emerges as a promising strategy to disrupt pathogenic fibroblast activity in IPF.

Data availability: The transcriptomic data was deposited into the GEO database under accession number GSE285246.

Ethics statement: The protocol was approved by the ethics committee of The Jikei University School of Medicine (#20-153) and Saitama Prefectural Cardiovascular and Respiratory Center (#2019044).

Author contributions: N. Watanabe, Y. Yamamoto and Y. Fujita conceived and designed the study. N. Watanabe, M. Yoshida, Y. Hirano, S. Matsubayashi and S. Fujimoto conducted the experiments. N. Watanabe, M. Yoshida, Y. Hirano and Y. Yamamoto performed data analysis. N. Watanabe, M. Yoshida, Y. Hirano, Y. Mori, K. Okamoto, S. Minagawa, K. Kuwano, J. Araya, Y. Yamamoto and Y. Fujita interpreted the data. T. Ishiguro, N. Takahashi, Y. Shimizu, N. Takayanagi and Y. Kawabata provided the patient samples. N. Watanabe, M. Yoshida, Y. Hirano, Y. Yamamoto and Y. Fujita wrote the manuscript. Y. Yamamoto and Y. Fujita supervised the study. All authors reviewed and edited the manuscript.

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